Kron's moving inductors

An induction motor started in Kron's frames I and II, and a parametric inductance pump against Mathieu's equation. CLAUDE.md §3a, §3l item 3; milestone M9c.1

kronspice simulation report · 2026-11-07 · transient 1.5 s, 0.5 ms step, RelTol 1e-5

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Summary

This is the milestone M9c.1 example for the N machine element (*device/machine.go*). One two-phase induction motor is started from rest twice: in Kron's holonomic frame I, where the rotor windings sit on the rotor and the inductances are trigonometric in the rotor angle θ, and in his non-holonomic rotor frame II(b), where the inductances are constant and speed voltages G·i·ω appear. The two runs must agree. Then the same element, with its coordinate driven at constant speed, is a time-varying inductance pumping an LC tank, and its growth is checked against Mathieu's equation.

- **The two frames agree.** Final speeds 313.568 rad/s (frame I) and 313.696 rad/s (frame II(b)), against synchronous 314.16 rad/s; over the whole start-up, speed within 1.1×10^{-4} and stator current within 5.8×10^{-4} of peak.
- **Both ledgers close:** supply 326.10 J = heat 227.29 J + stored 98.75 J (frame I, mismatch 9.8×10^{-5}); 326.21 J = 227.38 J + 98.83 J (frame II(b), mismatch -1.4×10^{-6}).
- **Speed terms appear only in the rotor rows** of the transformed tensor G′, as Kron's frame II requires; a frame that leaves θ in L is refused.
- **The pumped tank grows at Mathieu's rate:** 78.550 /s against $m\cdot\omega_0/4$ = 78.540 /s undamped, 39.273 /s against 39.270 /s at half damping, and decays at −78.571 /s against −78.540 /s when damping is twice the pump rate.
- **The pump pays for everything the tank gains:** pump work = tank gain + heat to 6×10^{-4}. A coefficient of performance computed from the capacitor's source alone, as in H16, omits that work.

The construction

The primitive machine

Kron's two-phase primitive machine has stator windings α, β and rotor windings a, b at angle θ, self-inductances Ls = Lr = 0.12 H and stator–rotor mutuals M·R(θ) with M = 0.11 H, so the inductance tensor is [[Ls·I, M·R(θ)], [$M\cdot R(\theta)^T$, Lr·I]], with R the 2×2 rotation. In kronspice this is a *circuit.MachineSpec* with a first harmonic in θ.

Frame I, holonomic

The element is solved with L(θ) as written: Kron's eq. 3, $e_\alpha = R_{\alpha\beta}\cdot i^\beta + L_{\alpha\beta}\cdot di^\beta/dt + [\beta\gamma, \alpha]\cdot i^\beta\cdot i^\gamma$, with the Christoffel symbols [βγ, α] of eq. 4 built from the metric (L(θ), J), and the torque $\tfrac{1}{2}\cdot i^T\cdot(\partial L/\partial\theta)\cdot i$ (*Equivalent Circuits of Electric Machinery*, 1951, Epilogue pp. 254–255). *kron.Christoffel* and *kron.Geodesic* reproduce the element's voltages and torque from those symbols to 10^{-12}.

Frame II(b), the rotor frame

kron.RotatingFrame applies the connection tensor C(θ) of *kron.RotorFrame* to the rotor axes (power invariant, e′ = $C^T\cdot e$). L′ = $C^T\cdot L\cdot C$ becomes constant and the rotor rows gain the speed matrix G = $C^T\cdot L\cdot\partial C/\partial\theta$, Kron's non-holonomic object, so that e = R·i + p(L·i) + G·i·pθ (p = d/dt) with torque $i^T\cdot G\cdot i$ (Epilogue eq. 9, p. 257). The program prints the transformed tensors:

```
L' = [[0.12 0 0.11 0], [0 0.12 0 0.11], [0.11 0 0.12 0], [0 0.11 0 0.12]]
G' = [[0 0 0 0], [0 0 0 0], [0 0.11 0 0.12], [-0.11 0 -0.12 0]]
```


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The motor circuit

part	value
supply	balanced two-phase 100 V, 50 Hz (sin and cos sources), 0.5 Ω per phase
rotor windings	shorted through 0.8 Ω
rotor inertia	0.002 kg·m², as a capacitor on the mechanical node w
friction load	0.2 mN·m per rad/s, as a 5 kΩ resistor on node w (angular velocity across, torque through)
synchronous speed	314.16 rad/s
transient	1.5 s, 0.5 ms step, RelTol 1e-5

The parametric pump

One winding with $L(t) = L0\cdot(1 + m\cdot\cos 2\omega_0 t)$, L0 = 1 mH, m = 0.05, made from a second harmonic in θ with the shaft turned at ω_0 by a voltage source on the mechanical node, so $\theta = \omega_0 t$ and the inductance varies at $2\omega_0$. It is in series with $C = 1/(\omega_0^2 L0)$ (1 V initial charge, f0 = 1 kHz) and a resistor $R = \text{damp}\cdot(m\cdot\omega_0/4)\cdot 2L0$, where damp is the damping R/(2L0) as a multiple of the pump rate. The tank runs 60 periods at 100 steps per period; the growth rate is the slope of log(peak) over periods 30 to 59.

Results

Start-up in the two frames

frame	final speed	points	supply	heat	stored (magnetic + kinetic)	ledger mismatch
I, holonomic, L(θ)	313.568 rad/s	7953	326.10 J	227.29 J	98.75 J	9.8×10^{-5}
II(b), rotor frame, G·i·ω	313.696 rad/s	8439	326.21 J	227.38 J	98.83 J	-1.4×10^{-6}

- The test holds the two final speeds within 10^{-3} of each other and the motor between 300 rad/s and synchronous (*analysis.TestMachineFrameTwin*).
- The ledger mismatch is the trapezoidal integration error over the sources' breakpoints.

The pumped tank against Mathieu

$\omega_0 = 2\pi\cdot 1$ kHz, m = 0.05, pump rate $m\cdot\omega_0/4 = 78.54$ /s.

damping R/2L0 (× pump rate)	growth, run	Mathieu $m\cdot\omega_0/4 - R/2L0$	pump work	heat
0	78.550 /s	78.540 /s	0.07876 J	5.1×10^{-10} J
0.5	39.273 /s	39.270 /s	1.396 mJ	0.707 mJ
2	−78.571 /s	−78.540 /s	4.38 µJ	17.1 µJ

- The test holds the growth within 1 % of Mathieu's rate for the first two rows (*analysis.TestMachineParametric*).
- The energy the tank gains enters through the pump's mechanical port: pump work = tank gain + heat, mismatch ≤ 6×10^{-4}.
- This is the standard account of parametric excitation (Mandelstam and Papalexi 1932; Kouřil and Vrba §10.2.2, whose averaged Mathieu exponent $\sigma = -\alpha + \sqrt{((m\omega_0/4)^2 - \Delta\omega^2)}$ reduces to the column above at zero detuning). It is the standard counterpart of the parametric circuits of H16 (*parametric_lcr*): the agent that varies L pays for everything the tank gains.

Exports

export/ holds the frame-I and frame-II motors (.tran 0.5m 1.5 uic) and the damping-0.5 parametric tank (.tran 10u 60m uic) as netlists with the N element and .model … machine, KiCad schematics and results sheets. The sheets show the DC operating point by MNA only: the machine is nonlinear, so Kron's linear torn solution is not drawn, and the torn Newton solve is checked in *kron*'s own tests.

netlist	elements, nodes	DC operating point (MNA)
kron_machine_motor_frame1.cir	9, 7	sa = 100 V; ia, ib, ra, rb, sb, w = 0
kron_machine_motor_frame2.cir	9, 7	sa = 100 V; ia, ib, ra, rb, sb, w = 0

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netlist	elements, nodes	DC operating point (MNA)
kron_machine_parametric.cir	4, 3	w = 6283.19 (shaft speed ω_0); a, b = 0

Validation

- Frame twin: speeds agree to 10^{-3}, near synchronous; speed within 1.1×10^{-4} and stator current within 5.8×10^{-4} of peak over the start-up (doc/M9c1.md).
- Christoffel form (Kron 1951 eqs. 3–4) equals the element to 10^{-12} (*kron* tests).
- Mathieu growth within 1 %; pump-work ledger mismatch ≤ 6×10^{-4}.

Limits

- The machine has a first-harmonic winding distribution only (the motor) or a second harmonic only (the pump).
- The growth rate is fitted from peaks over periods 30 to 59 at 100 steps per period; the fit and the trapezoidal integrator set the small differences from Mathieu's rate.
- The example has no run_output.txt or figures; numbers are those recorded in its README and export sheets.

Reproducing the results

```
go run ./examples/kron_machine
go test ./examples/kron_machine
KRONSPICE_EXPORT=1 go test ./examples/kron_machine -run TestExport   # netlists, KiCad, results sheets
```

Sources: G. Kron, Equivalent Circuits of Electric Machinery, Wiley, 1951, Epilogue pp. 253–265; G. Kron, J. Math. Phys. 13, 103–194 (1934); F. Kouřil and K. Vrba, Non-linear and Parametric Circuits, 1988, §10.2.2; L. Mandelstam and N. Papalexi, Z. Phys. 73, 223 (1932); doc/M9c1.md.

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